```
defparam ram512x8_inst.INIT_6 =
256'h0000000000000000000000000000000000000000000000000000000000000000;
defparam ram512x8_inst.INIT_7 =
256'h0000000000000000000000000000000000000000000000000000000000000000;
defparam ram512x8_inst.INIT_8 =
256'h0000000000000000000000000000000000000000000000000000000000000000;
defparam ram512x8_inst.INIT_9 =
256'h0000000000000000000000000000000000000000000000000000000000000000;
defparam ram512x8_inst.INIT_A =
256'h0000000000000000000000000000000000000000000000000000000000000000;
defparam ram512x8_inst.INIT_B =
256'h0000000000000000000000000000000000000000000000000000000000000000;
defparam ram512x8_inst.INIT_C =
256'h0000000000000000000000000000000000000000000000000000000000000000;
defparam ram512x8_inst.INIT_D =
256'h0000000000000000000000000000000000000000000000000000000000000000;
defparam ram512x8_inst.INIT_E =
256'h0000000000000000000000000000000000000000000000000000000000000000;
defparam ram512x8_inst.INIT_F =
256'h0000000000000000000000000000000000000000000000000000000000000000;
```

VHDL Instantiation:

```
ram512x8_inst :  SB_RAM512x8
generic map (
INIT_0 =>
X"0000000000000000000000000000000000000000000000000000000000000000",
INIT_1 =>
X"0000000000000000000000000000000000000000000000000000000000000000",
INIT_2 =>
X"0000000000000000000000000000000000000000000000000000000000000000",
INIT_3 =>
X"0000000000000000000000000000000000000000000000000000000000000000",
INIT_4 =>
X"0000000000000000000000000000000000000000000000000000000000000000",
INIT_5 =>
X"0000000000000000000000000000000000000000000000000000000000000000",
INIT_6 =>
X"0000000000000000000000000000000000000000000000000000000000000000",
INIT_7 =>
X"0000000000000000000000000000000000000000000000000000000000000000",
INIT_8 =>
X"0000000000000000000000000000000000000000000000000000000000000000",
INIT_9 =>
X"0000000000000000000000000000000000000000000000000000000000000000",
INIT_A =>
X"0000000000000000000000000000000000000000000000000000000000000000",
INIT_B =>
X"0000000000000000000000000000000000000000000000000000000000000000",
INIT_C =>
X"0000000000000000000000000000000000000000000000000000000000000000",
INIT_D =>
X"0000000000000000000000000000000000000000000000000000000000000000",
INIT_E =>
X"0000000000000000000000000000000000000000000000000000000000000000",
INIT_F => X"0000000000000000000000000000000000000000000000000000000000000000"
      )
port map (
        RDATA => RDATA_c,
        RADDR => RADDR_c,
        RCLK => RCLK_c,
        RCLKE => RCLKE_c,
        RE => RE_c,
```

```
          WADDR => WADDR_c,
          WCLK=> WCLK_c,
          WCLKE => WCLKE_c,
          WDATA => WDATA_c,
          WE => WE_c
        );
```

SB_RAM512x8NR

SB_RAM512x8NR // Negative edged Read Clock – i.e. RCLKN
(RDATA, RCLKN, RCLKE, RE, RADDR, WCLK, WCLKE, WE, WADDR, MASK, WDATA);

Verilog Instantiation:

```
SB_RAM512x8NR   ram512x8nr_inst (
      .RDATA(RDATA_c[7:0]),
      .RADDR(RADDR_c[8:0]),
      .RCLKN(RCLKN_c),
      .RCLKE(RCLKE_c),
      .RE(RE_c),
      .WADDR(WADDR_c[8:0]),
      .WCLK(WCLK_c),
      .WCLKE(WCLKE_c),
      .WDATA(WDATA_c[7:0]),
      .WE(WE_c)
);

defparam ram512x8nr_inst.INIT_0 =
256'h0000000000000000000000000000000000000000000000000000000000000000;
defparam ram512x8nr_inst.INIT_1 =
256'h0000000000000000000000000000000000000000000000000000000000000000;
defparam ram512x8nr_inst.INIT_2 =
256'h0000000000000000000000000000000000000000000000000000000000000000;
defparam ram512x8nr_inst.INIT_3 =
256'h0000000000000000000000000000000000000000000000000000000000000000;
defparam ram512x8nr_inst.INIT_4 =
256'h0000000000000000000000000000000000000000000000000000000000000000;
defparam ram512x8nr_inst.INIT_5 =
256'h0000000000000000000000000000000000000000000000000000000000000000;
defparam ram512x8nr_inst.INIT_6 =
256'h0000000000000000000000000000000000000000000000000000000000000000;
defparam ram512x8nr_inst.INIT_7 =
256'h0000000000000000000000000000000000000000000000000000000000000000;
defparam ram512x8nr_inst.INIT_8 =
256'h0000000000000000000000000000000000000000000000000000000000000000;
defparam ram512x8nr_inst.INIT_9 =
256'h0000000000000000000000000000000000000000000000000000000000000000;
defparam ram512x8nr_inst.INIT_A =
256'h0000000000000000000000000000000000000000000000000000000000000000;
defparam ram512x8nr_inst.INIT_B =
256'h0000000000000000000000000000000000000000000000000000000000000000;
defparam ram512x8nr_inst.INIT_C =
256'h0000000000000000000000000000000000000000000000000000000000000000;
defparam ram512x8nr_inst.INIT_D =
256'h0000000000000000000000000000000000000000000000000000000000000000;
defparam ram512x8nr_inst.INIT_E =
256'h0000000000000000000000000000000000000000000000000000000000000000;
defparam ram512x8nr_inst.INIT_F =
256'h0000000000000000000000000000000000000000000000000000000000000000;
```

VHDL Instantiation:

```
ram512x8nr_inst:  SB_RAM512x8NR
generic map (
INIT_0 =>
X"0000000000000000000000000000000000000000000000000000000000000000",
INIT_1 =>
X"0000000000000000000000000000000000000000000000000000000000000000",
INIT_2 =>
X"0000000000000000000000000000000000000000000000000000000000000000",
INIT_3 =>
X"0000000000000000000000000000000000000000000000000000000000000000",
INIT_4 =>
X"0000000000000000000000000000000000000000000000000000000000000000",
INIT_5 =>
X"0000000000000000000000000000000000000000000000000000000000000000",
INIT_6 =>
X"0000000000000000000000000000000000000000000000000000000000000000",
INIT_7 =>
X"0000000000000000000000000000000000000000000000000000000000000000",
INIT_8 =>
X"0000000000000000000000000000000000000000000000000000000000000000",
INIT_9 =>
X"0000000000000000000000000000000000000000000000000000000000000000",
INIT_A =>
X"0000000000000000000000000000000000000000000000000000000000000000",
INIT_B =>
X"0000000000000000000000000000000000000000000000000000000000000000",
INIT_C =>
X"0000000000000000000000000000000000000000000000000000000000000000",
INIT_D =>
X"0000000000000000000000000000000000000000000000000000000000000000",
INIT_E =>
X"0000000000000000000000000000000000000000000000000000000000000000",
INIT_F => X"0000000000000000000000000000000000000000000000000000000000000000"
      )
port map (
        RDATA => RDATA_c,
        RADDR => RADDR_c,
        RCLKN => RCLKN_c,
        RCLKE => RCLKE_c,
        RE => RE_c,
        WADDR => WADDR_c,
        WCLK=> WCLK_c,
        WCLKE => WCLKE_c,
        WDATA => WDATA_c,
        WE => WE_c
      );
```

SB_RAM512x8NW

SB_RAM512x8NW **// Negative edged Write Clock – i.e. WCLKN**
(RDATA, RCLK, RCLKE, RE, RADDR, WCLKN, WCLKE, WE, WADDR, MASK, WDATA);

Verilog Instantiation:

```
SB_RAM512x8NW   ram512x8nw_inst (
      .RDATA(RDATA_c[7:0]),
      .RADDR(RADDR_c[8:0]),
```

```
      .RCLK(RCLK_c),
      .RCLKE(RCLKE_c),
      .RE(RE_c),
      .WADDR(WADDR_c[8:0]),
      .WCLKN(WCLKN_c),
      .WCLKE(WCLKE_c),
      .WDATA(WDATA_c[7:0]),
      .WE(WE_c)
);

defparam ram512x8nw_inst.INIT_0 =
256'h0000000000000000000000000000000000000000000000000000000000000000;
defparam ram512x8nw_inst.INIT_1 =
256'h0000000000000000000000000000000000000000000000000000000000000000;
defparam ram512x8nw_inst.INIT_2 =
256'h0000000000000000000000000000000000000000000000000000000000000000;
defparam ram512x8nw_inst.INIT_3 =
256'h0000000000000000000000000000000000000000000000000000000000000000;
defparam ram512x8nw_inst.INIT_4 =
256'h0000000000000000000000000000000000000000000000000000000000000000;
defparam ram512x8nw_inst.INIT_5 =
256'h0000000000000000000000000000000000000000000000000000000000000000;
defparam ram512x8nw_inst.INIT_6 =
256'h0000000000000000000000000000000000000000000000000000000000000000;
defparam ram512x8nw_inst.INIT_7 =
256'h0000000000000000000000000000000000000000000000000000000000000000;
defparam ram512x8nw_inst.INIT_8 =
256'h0000000000000000000000000000000000000000000000000000000000000000;
defparam ram512x8nw_inst.INIT_9 =
256'h0000000000000000000000000000000000000000000000000000000000000000;
defparam ram512x8nw_inst.INIT_A =
256'h0000000000000000000000000000000000000000000000000000000000000000;
defparam ram512x8nw_inst.INIT_B =
256'h0000000000000000000000000000000000000000000000000000000000000000;
defparam ram512x8nw_inst.INIT_C =
256'h0000000000000000000000000000000000000000000000000000000000000000;
defparam ram512x8nw_inst.INIT_D =
256'h0000000000000000000000000000000000000000000000000000000000000000;
defparam ram512x8nw_inst.INIT_E =
256'h0000000000000000000000000000000000000000000000000000000000000000;
defparam ram512x8nw_inst.INIT_F =
256'h0000000000000000000000000000000000000000000000000000000000000000;
```

VHDL Instantiation:

```
ram512x8nw_inst:  SB_RAM512x8NW
generic map (
INIT_0 =>
X"0000000000000000000000000000000000000000000000000000000000000000",
INIT_1 =>
X"0000000000000000000000000000000000000000000000000000000000000000",
INIT_2 =>
X"0000000000000000000000000000000000000000000000000000000000000000",
INIT_3 =>
X"0000000000000000000000000000000000000000000000000000000000000000",
INIT_4 =>
X"0000000000000000000000000000000000000000000000000000000000000000",
INIT_5 =>
X"0000000000000000000000000000000000000000000000000000000000000000",
INIT_6 =>
X"0000000000000000000000000000000000000000000000000000000000000000",
INIT_7 =>
X"0000000000000000000000000000000000000000000000000000000000000000",
```

```
INIT_8 =>
X"0000000000000000000000000000000000000000000000000000000000000000",
INIT_9 =>
X"0000000000000000000000000000000000000000000000000000000000000000",
INIT_A =>
X"0000000000000000000000000000000000000000000000000000000000000000",
INIT_B =>
X"0000000000000000000000000000000000000000000000000000000000000000",
INIT_C =>
X"0000000000000000000000000000000000000000000000000000000000000000",
INIT_D =>
X"0000000000000000000000000000000000000000000000000000000000000000",
INIT_E =>
X"0000000000000000000000000000000000000000000000000000000000000000",
INIT_F => X"0000000000000000000000000000000000000000000000000000000000000000"
      )
port map (
        RDATA => RDATA_c,
        RADDR => RADDR_c,
        RCLK => RCLK_c,
        RCLKE => RCLKE_c,
        RE => RE_c,
        WADDR => WADDR_c,
        WCLKN=> WCLKN_c,
        WCLKE => WCLKE_c,
        WDATA => WDATA_c,
        WE => WE_c
      );
```

SB_RAM512x8NRNW

SB_RAM512x8NRNW **// Negative edged Read and Write – i.e. RCLKN WRCKLN**
(RDATA, RCLKN, RCLKE, RE, RADDR, WCLKN, WCLKE, WE, WADDR, MASK, WDATA);
Verilog Instantiation:

```
SB_RAM512x8NRNW   ram512x8nrnw_inst (
      .RDATA(RDATA_c[7:0]),
      .RADDR(RADDR_c[8:0]),
      .RCLKN(RCLK_c),
      .RCLKE(RCLKE_c),
      .RE(RE_c),
      .WADDR(WADDR_c[8:0]),
      .WCLKN(WCLK_c),
      .WCLKE(WCLKE_c),
      .WDATA(WDATA_c[7:0]),
      .WE(WE_c)
);

defparam ram512x8nrnw_inst.INIT_0 =
256'h0000000000000000000000000000000000000000000000000000000000000000;
defparam ram512x8nrnw_inst.INIT_1 =
256'h0000000000000000000000000000000000000000000000000000000000000000;
defparam ram512x8nrnw_inst.INIT_2 =
256'h0000000000000000000000000000000000000000000000000000000000000000;
defparam ram512x8nrnw_inst.INIT_3 =
256'h0000000000000000000000000000000000000000000000000000000000000000;
defparam ram512x8nrnw_inst.INIT_4 =
256'h0000000000000000000000000000000000000000000000000000000000000000;
```

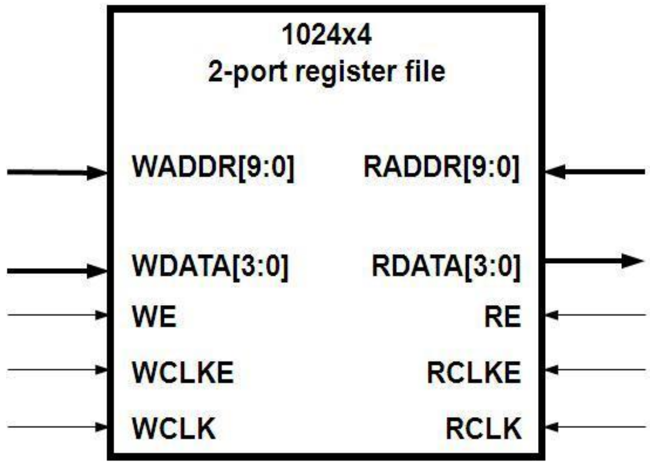
```
defparam ram512x8nrnw_inst.INIT_5 =
256'h0000000000000000000000000000000000000000000000000000000000000000;
defparam ram512x8nrnw_inst.INIT_6 =
256'h0000000000000000000000000000000000000000000000000000000000000000;
defparam ram512x8nrnw_inst.INIT_7 =
256'h0000000000000000000000000000000000000000000000000000000000000000;
defparam ram512x8nrnw_inst.INIT_8 =
256'h0000000000000000000000000000000000000000000000000000000000000000;
defparam ram512x8nrnw_inst.INIT_9 =
256'h0000000000000000000000000000000000000000000000000000000000000000;
defparam ram512x8nrnw_inst.INIT_A =
256'h0000000000000000000000000000000000000000000000000000000000000000;
defparam ram512x8nrnw_inst.INIT_B =
256'h0000000000000000000000000000000000000000000000000000000000000000;
defparam ram512x8nrnw_inst.INIT_C =
256'h0000000000000000000000000000000000000000000000000000000000000000;
defparam ram512x8nrnw_inst.INIT_D =
256'h0000000000000000000000000000000000000000000000000000000000000000;
defparam ram512x8nrnw_inst.INIT_E =
256'h0000000000000000000000000000000000000000000000000000000000000000;
defparam ram512x8nrnw_inst.INIT_F =
256'h0000000000000000000000000000000000000000000000000000000000000000;
```

VHDL Instantiation:

```
ram512x8nrnw_inst:  SB_RAM512x8NRNW
generic map (
INIT_0 =>
X"0000000000000000000000000000000000000000000000000000000000000000",
INIT_1 =>
X"0000000000000000000000000000000000000000000000000000000000000000",
INIT_2 =>
X"0000000000000000000000000000000000000000000000000000000000000000",
INIT_3 =>
X"0000000000000000000000000000000000000000000000000000000000000000",
INIT_4 =>
X"0000000000000000000000000000000000000000000000000000000000000000",
INIT_5 =>
X"0000000000000000000000000000000000000000000000000000000000000000",
INIT_6 =>
X"0000000000000000000000000000000000000000000000000000000000000000",
INIT_7 =>
X"0000000000000000000000000000000000000000000000000000000000000000",
INIT_8 =>
X"0000000000000000000000000000000000000000000000000000000000000000",
INIT_9 =>
X"0000000000000000000000000000000000000000000000000000000000000000",
INIT_A =>
X"0000000000000000000000000000000000000000000000000000000000000000",
INIT_B =>
X"0000000000000000000000000000000000000000000000000000000000000000",
INIT_C =>
X"0000000000000000000000000000000000000000000000000000000000000000",
INIT_D =>
X"0000000000000000000000000000000000000000000000000000000000000000",
INIT_E =>
X"0000000000000000000000000000000000000000000000000000000000000000",
INIT_F => X"0000000000000000000000000000000000000000000000000000000000000000"
      )
port map (
        RDATA => RDATA_c,
        RADDR => RADDR_c,
        RCLKN => RCLKN_c,
```

```
    RCLKE => RCLKE_c,
    RE => RE_c,
    WADDR => WADDR_c,
    WCLKN=> WCLKN_c,
    WCLKE => WCLKE_c,
    WDATA => WDATA_c,
    WE => WE_c
  );
```

SB_RAM1024x4



SB_RAM1024x4

The following modules are the complete list of SB_RAM1024x4 based primitives

SB_RAM1024x4

SB_RAM1024x4 **//Posedge clock RCLK WCLK**
(RDATA, RCLK, RCLKE, RE, RADDR, WCLK, WCLKE, WE, WADDR, MASK, WDATA);

Verilog Instantiation:

```
SB_RAM1024x4   ram1024x4_inst (
       .RDATA(RDATA_c[3:0]),
       .RADDR(RADDR_c[9:0]),
       .RCLK(RCLK_c),
       .RCLKE(RCLKE_c),
       .RE(RE_c),
       .WADDR(WADDR_c[3:0]),
       .WCLK(WCLK_c),
       .WCLKE(WCLKE_c),
       .WDATA(WDATA_c[9:0]),
       .WE(WE_c)
);
defparam ram1024x4_inst.INIT_0 =
256'h0000000000000000000000000000000000000000000000000000000000000000;
defparam ram1024x4_inst.INIT_1 =
256'h0000000000000000000000000000000000000000000000000000000000000000;
defparam ram1024x4_inst.INIT_2 =
256'h0000000000000000000000000000000000000000000000000000000000000000;
defparam ram1024x4_inst.INIT_3 =
256'h0000000000000000000000000000000000000000000000000000000000000000;
defparam ram1024x4_inst.INIT_4 =
256'h0000000000000000000000000000000000000000000000000000000000000000;
```

```
defparam ram1024x4_inst.INIT_5 =
256'h0000000000000000000000000000000000000000000000000000000000000000;
defparam ram1024x4_inst.INIT_6 =
256'h0000000000000000000000000000000000000000000000000000000000000000;
defparam ram1024x4_inst.INIT_7 =
256'h0000000000000000000000000000000000000000000000000000000000000000;
defparam ram1024x4_inst.INIT_8 =
256'h0000000000000000000000000000000000000000000000000000000000000000;
defparam ram1024x4_inst.INIT_9 =
256'h0000000000000000000000000000000000000000000000000000000000000000;
defparam ram1024x4_inst.INIT_A =
256'h0000000000000000000000000000000000000000000000000000000000000000;
defparam ram1024x4_inst.INIT_B =
256'h0000000000000000000000000000000000000000000000000000000000000000;
defparam ram1024x4_inst.INIT_C =
256'h0000000000000000000000000000000000000000000000000000000000000000;
defparam ram1024x4_inst.INIT_D =
256'h0000000000000000000000000000000000000000000000000000000000000000;
defparam ram1024x4_inst.INIT_E =
256'h0000000000000000000000000000000000000000000000000000000000000000;
defparam ram1024x4_inst.INIT_F =
256'h0000000000000000000000000000000000000000000000000000000000000000;
```

VHDL Instantiation:

```
Ram1024x4_inst:  SB_RAM1024x4
generic map (
INIT_0 =>
X"0000000000000000000000000000000000000000000000000000000000000000",
INIT_1 =>
X"0000000000000000000000000000000000000000000000000000000000000000",
INIT_2 =>
X"0000000000000000000000000000000000000000000000000000000000000000",
INIT_3 =>
X"0000000000000000000000000000000000000000000000000000000000000000",
INIT_4 =>
X"0000000000000000000000000000000000000000000000000000000000000000",
INIT_5 =>
X"0000000000000000000000000000000000000000000000000000000000000000",
INIT_6 =>
X"0000000000000000000000000000000000000000000000000000000000000000",
INIT_7 =>
X"0000000000000000000000000000000000000000000000000000000000000000",
INIT_8 =>
X"0000000000000000000000000000000000000000000000000000000000000000",
INIT_9 =>
X"0000000000000000000000000000000000000000000000000000000000000000",
INIT_A =>
X"0000000000000000000000000000000000000000000000000000000000000000",
INIT_B =>
X"0000000000000000000000000000000000000000000000000000000000000000",
INIT_C =>
X"0000000000000000000000000000000000000000000000000000000000000000",
INIT_D =>
X"0000000000000000000000000000000000000000000000000000000000000000",
INIT_E =>
X"0000000000000000000000000000000000000000000000000000000000000000",
INIT_F => X"0000000000000000000000000000000000000000000000000000000000000000"
      )
port map (
        RDATA => RDATA_c,
        RADDR => RADDR_c,
        RCLK => RCLK_c,
```

```
        RCLKE => RCLKE_c,
        RE => RE_c,
        WADDR => WADDR_c,
        WCLK=> WCLK_c,
        WCLKE => WCLKE_c,
        WDATA => WDATA_c,
        WE => WE_c
      );
```

SB_RAM1024x4NR

SB_RAM1024x4NR **// Negative edged Read Clock – i.e. RCLKN**
(RDATA, RCLKN, RCLKE, RE, RADDR, WCLK, WCLKE, WE, WADDR, MASK, WDATA);

Verilog Instantiation:

```
SB_RAM1024x4NR   ram1024x4nr_inst (
      .RDATA(RDATA_c[3:0]),
      .RADDR(RADDR_c[9:0]),
      .RCLKN(RCLKN_c),
      .RCLKE(RCLKE_c),
      .RE(RE_c),
      .WADDR(WADDR_c[3:0]),
      .WCLK(WCLK_c),
      .WCLKE(WCLKE_c),
      .WDATA(WDATA_c[9:0]),
      .WE(WE_c)
);
defparam ram1024x4nr_inst.INIT_0 =
256'h0000000000000000000000000000000000000000000000000000000000000000;
defparam ram1024x4nr_inst.INIT_1 =
256'h0000000000000000000000000000000000000000000000000000000000000000;
defparam ram1024x4nr_inst.INIT_2 =
256'h0000000000000000000000000000000000000000000000000000000000000000;
defparam ram1024x4nr_inst.INIT_3 =
256'h0000000000000000000000000000000000000000000000000000000000000000;
defparam ram1024x4nr_inst.INIT_4 =
256'h0000000000000000000000000000000000000000000000000000000000000000;
defparam ram1024x4nr_inst.INIT_5 =
256'h0000000000000000000000000000000000000000000000000000000000000000;
defparam ram1024x4nr_inst.INIT_6 =
256'h0000000000000000000000000000000000000000000000000000000000000000;
defparam ram1024x4nr_inst.INIT_7 =
256'h0000000000000000000000000000000000000000000000000000000000000000;
defparam ram1024x4nr_inst.INIT_8 =
256'h0000000000000000000000000000000000000000000000000000000000000000;
defparam ram1024x4nr_inst.INIT_9 =
256'h0000000000000000000000000000000000000000000000000000000000000000;
defparam ram1024x4nr_inst.INIT_A =
256'h0000000000000000000000000000000000000000000000000000000000000000;
defparam ram1024x4nr_inst.INIT_B =
256'h0000000000000000000000000000000000000000000000000000000000000000;
defparam ram1024x4nr_inst.INIT_C =
256'h0000000000000000000000000000000000000000000000000000000000000000;
defparam ram1024x4nr_inst.INIT_D =
256'h0000000000000000000000000000000000000000000000000000000000000000;
defparam ram1024x4nr_inst.INIT_E =
256'h0000000000000000000000000000000000000000000000000000000000000000;
defparam ram1024x4nr_inst.INIT_F =
256'h0000000000000000000000000000000000000000000000000000000000000000;
```